

# LM1 Interoperate Investments.

## Investment Categories

债权及未产生大影响的股权

长期股权投资

|                        | Financial Assets         | 联营 Associates         | 合营 Joint Ventures | Business 子嗣 Combination               |
|------------------------|--------------------------|-----------------------|-------------------|---------------------------------------|
| Degree of Influence    | no significant influence | significant influence | shared control    | control                               |
| Percentage of Interest | < 20%                    | 20% - 50%             | varies            | > 50%                                 |
| Term of Investee       |                          | associate             |                   | subsidiary                            |
| Accounting Treatment   | AMC<br>FVOCI<br>FVPL     | Equity Method         | Equity Method     | Consolidation /<br>Acquisition Method |

只是 guideline 更看实质影响

[ Percentage of ownership ]

|           |                       |
|-----------|-----------------------|
| < 20%     | Lack of Influence     |
| 20% ~ 50% | Significant Influence |
| > 50%     | Control               |

direct 与 indirect 合计

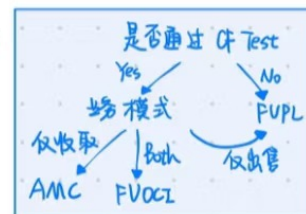
## Financial Asset

1 分类 2 原则 { Cash Flow Characteristic Test : 未来CF是否基于本金及利息的合同现金流?

Business Model Test : 收取合同CF / 出售 Financial Asset / Both

3 分类 { AMC : 通过 CF Test 且目的仅收取合同CF  
FVOCI : 通过 CF Test 且目的 Both  
FVPL : 其余所有 (衍生品一定 FVPL) 兜底

AMC 只有债券  
股票一定在 FVOCI or FVPL



2 指定 { FVPL option: 指定为 FVPL 以避免 accounting mismatch. (债)

FVOCI: 指定非交易性 股权投资为 FVOCI. (股)

一经指定, 不得改变!

## 2 Measurement

只有 Div Income 进 I/S. (R/E).

Impairment: Expected Loss Model

|                  | FV through P/L                                                                                                        | FV through OCI                                                                                                                                     | Amortized cost                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Balance sheet    | Fair value                                                                                                            | <ul style="list-style-type: none"> <li>Fair value</li> <li>Unrealized G/L -&gt; OCI (I/S when realized)</li> </ul> 例外: 在 IFRS 下, 投资外国债券的外汇变动计入 I/S | Amortized cost<br>不认 Unrealized G/L                                              |
| Income statement | <ul style="list-style-type: none"> <li>Interest</li> <li>Dividend</li> <li>Realized or unrealized G/L</li> </ul> 浮盈浮亏 | <ul style="list-style-type: none"> <li>Interest</li> <li>Dividend</li> <li>Realized G/L</li> </ul>                                                 | <ul style="list-style-type: none"> <li>Interest</li> <li>Realized G/L</li> </ul> |

平价

$$I = \text{coupon} \\ A = S$$

溢价

$$I < \text{coupon} \\ A < S$$

折价

$$I > \text{coupon} \\ A > S$$

若 Par Value 发行  
均始终保持面额

Amortized cost: B/S 中的 AC  
BASE 法则.  
Beginning + 应收 - 实收 = Ending.  
Inv. yr. Coupon

3 类 Int. Income

$$= \text{Beginning Investment} \times r \text{ 都进 I/S.}$$

## ② Reclassification (不重要, 反毛即可)

Financial Asset  $\begin{cases} \text{股} - \text{不允许重分类, 仍是} \begin{cases} \text{FVPL} \\ \text{指定FVOCI} \end{cases} \\ \text{债} - \text{业务模式重大变动可重分类} \end{cases}$

金融负债、衍生品 - 经确认不可重分类  $\begin{cases} \text{AMC} \\ \text{FVOCI} \\ \text{FVPL} \end{cases}$  指定FVPL不可重分类

Reclassification Date: 业务模式变更后下一个会计周期的第一天、未来适用法

## Investment in Associates 联营 (associate or investee).

direct, indirect 合计, 且判断时  $\begin{cases} \text{IFRS: current 可行权的各类 option 以股合计, 记账以事实记} \\ \text{GAAP: option 不算.} \end{cases}$

## ★ ① Equity Method = one-line consolidation 单项合并

### ① Basic Principal:

初始: 长投 at cost (FV).

Then  $\begin{cases} \text{investee's net income 按比例记在 investor's 长投, RE 相应增长} \\ \text{investee 分红理解为退款, I/S 不处理, B/S 中: Cash +, 长投 -} \end{cases}$   $\therefore \text{investee 所有者权益降低了!}$

若 investee 亏钱, 长投以 0 为下限, 其余进入备查簿

公式: BASE 法则.

B/S:  $\text{Inv in Associate} = \text{Cost} + \text{NI} \times \% - \text{Div} \times \% \leftarrow \text{增加额外资产折旧! 因为NI要调成FV}$   
I/S:  $\text{Inv Income} = \text{NI} \times \%$

### ② Excess Purchase Price:

PP  $\begin{cases} \text{excess PP} \\ \text{FV} \end{cases} \begin{cases} \text{goodwill} \rightarrow \text{not amortized, review for impairment. 不单独列于报表, 隐含在长投中} \\ \text{FV appre} \rightarrow \text{FV - BV, Amortization of FV Appreciation. } \frac{\text{FV - BV}}{\text{year}} \\ \text{BV} \end{cases}$   $\therefore \text{是对长投进行减值测试}$

BV of net identifiable asset = Equity BV  
可辨认资产与负债差额

公式:

$\text{Cost} = \text{Purchase Price} = \text{B.V} + \text{F.V appre} + \text{Goodwill}$   
 $\text{Goodwill} = \text{PP} - \text{FV}$

### ③ Impairment of 长投

x temporary FV 高时常做减值测试, 一经计提减值不可退回.

IFRS:  $\text{Carrying Value} - \text{Recoverable Amount}$  (disposal value, value in use 高者).  
GAAP:  $\text{Carrying Value} - \text{FV}$   
BV of Identifiable Net Asset



#### ④ Transactions with associate.

NI调整未实现内部交易损益, 实现后再加回, 不重复操作

upstream 逆流: investee 卖给 investor  
downstream 顺流: investor 卖给 investee

顺流、逆流一样处理, 无论谁有 unrealized 都减!

未出售给独立第三方前/未消耗都是 unrealized!  
若是损失, 均认定为 realized!

NI 剔除 unrealized Profits

#### ⑤ FU option 公允价值选择权.

长投确认当天 < IFRS: 部分 investor 机构, 有权利选择  
GAAP: 全部机构

### Consolidation

#### ① Combination

控股合并 Acquisition:  $A+B=A+B$  → 只有控股合并编制合并报表  
吸收合并 Merger:  $A+B=A$   
新设合并 Combination:  $A+B=C$

同控: 权益结合法

非同控: 购买法 Acquisition Method.

#### ② Consolidation Process

##### ① B/S 合并思路

A公司(BV) 并 B公司(CV). 一定 100% 合并  
A与A合并, L与L合并, E不合并 → A调下 Minority Interest.

溢价部分单独计算 + 下 Partial Goodwill = 收购部分 Goodwill  
Full Goodwill = Partial Goodwill ÷ 收购%

Full ↔ Partial 若不平, 调 Minority Interest  
Full Goodwill → MI = 溢价 × 少数%  
Partial Goodwill → MI = PU × 少数%

只有 Consolidation 有 MI.  
Equity Proportionate 无 MI

##### ② I/S 合并思路

将 B公司 BV 调为 FU. 简单加总 (Inventory 导致 COGS 调整, F/A, I/A 导致 D&A 调整)

### Goodwill

#### ① 计算: 先 Partial 后 Full

② Impairment: IFRS: BV vs. Recoverable Amount (of reporting unit include GW) 超出部分减值, 到 0 为止 / 减 Asset  
GAAP: step 1: BV > FU 减值.  
step 2: BV - Implied FU.

#### Equity Method → Consolidation.

NI 不变! Equity ↑, ROE ↓.  
Asset ↑, ROA ↓.  
Sales ↑, net margin ↓

Equity Method 财务指标好看

pooling vs. purchase.

ROA, ROE, Profit margin 都更高.  
Historical Book Value, A-L 更小.

### SPE

IFRS: 必合并 → 不独立, 依赖母公司.  
US GAAP: VIE 合并  
not VIE 不合并 (过去).

## LM2: Employee Compensation

分类 DC VS DB

看谁承担风险 DC Plan 员工承担

PBO (PUBO) 计算

累计福利单位折现



基于退休前工资算 PMT

① 2种计算方法

退休工资 = Current annual salary  $\times (1+r)^{\text{year to retirement}-1}$

① PMT N. I/Y. FV=0  $\rightarrow$  PV  $PV \div (1+r)^n = PBO$

② Annual unit credit per service year (即 养老金退休时点 PV / 总年份)

PMT (only work 1 year) N. I/Y. FV=0  $\rightarrow$  PV 工作X年 PV 乘X 折现

② ABO

基于当前工资算 PMT

PBO: ABO = 退休前比当前工资倍数

B/S 核算 (Base 法则)

Funded status = Plan assets - PBO

| Plan Assets                                                                    |
|--------------------------------------------------------------------------------|
| Fair value at the end of the year<br>= Fair value at the beginning of the year |
| + Employer contributions 公司投钱                                                  |
| + Actual return 投资回报                                                           |
| - Benefit paid to employee 公司发钱                                                |

| PBO = PVC (养老金)                                                            |
|----------------------------------------------------------------------------|
| PBO at the end of the year<br>= PBO at the beginning of the year           |
| + Current service cost $+1\% + 1\%$ = PV ( $\Delta$ 养老金)                   |
| + Interest costs $= PBO_{\text{begin}} \times r$ = unit credit 折现          |
| + Past service cost (plan amendments during the year) 过去服务成本 $1\% - 1.5\%$ |
| + Actuarial losses                                                         |
| - Actuarial gains during the year > 假设改变                                   |
| - Benefit paid to employee 公司发钱                                            |

Plan asset > PBO  $\rightarrow$  Overfunded  
Plan asset < PBO  $\rightarrow$  Underfunded

$\rightarrow$  是资产时看有无超过 Ceiling

I/S 核算 TPPC

IFRS

|     |                                                           |               |
|-----|-----------------------------------------------------------|---------------|
| PIL | Current Service Cost                                      | Remeasurement |
|     | Past Service Cost                                         |               |
|     | Int Cost = $PBO_{\text{begin}} \times r$                  |               |
| OCI | - Int Income = $PA_{\text{begin}} \times r$ discount rate |               |
|     | Actuarial G/L 精算假设<br>Actual Return - Int Income          |               |

US GAAP

|     |                                                      |            |
|-----|------------------------------------------------------|------------|
| PIL | Current Service Cost                                 | 根据剩余工作年限摊销 |
|     | Int Cost = $PBO_{\text{begin}} \times r$             |            |
|     | - Expected Return = Asset $\times$ ECR               |            |
| OCI | Amortization of { Past Service Cost<br>Actuarial G/L |            |
|     | Unamortized { Past Service Cost<br>Actuarial G/L     |            |
|     |                                                      |            |

假设条件变化 r/g/ECR 方向判断即可

- $r \uparrow$  < B/S: PBO  $\downarrow$  Funded Status  $\uparrow$   
I/S: C.S.C  $\downarrow$  Int Cost =  $PBO_{\text{begin}} \times r$   $\downarrow$
- $g \downarrow \rightarrow$  退休前工资  $\downarrow$  < B/S: PBO  $\downarrow$  F.S  $\uparrow$   
I/S: C.S.C  $\downarrow$  Int Cost  $\downarrow$
- ECR  $\uparrow$  US GAAP < I/S: Pension Expense  $\downarrow$   
B/S: PBO F.S unchanged

| Assumption                            | Impact on Balance Sheet                                               | Impact on Periodic Cost in Income Statements                                                                | Impact on Total Periodic Pension Cost          | Conservative or aggressive assumption |
|---------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------|
| Higher Discount Rate                  | Lower obligation                                                      | Periodic pension costs will typically be lower, because of lower opening obligation and lower service costs | Lower TPPC, because of higher funded status    | Aggressive                            |
| Lower Rate of Compensation Growth     | Lower obligation                                                      | Lower periodic pension expense, because of lower service costs                                              | Lower TPPC, because of higher funded status    | Aggressive                            |
| Higher Expected Return on Plan Assets | No effect, because fair value of plan assets is used on balance sheet | Not applicable for IFRS; Lower periodic pension expense under US GAAP (US 中假设可调整)                           | No effect, because no effects on funded status | Aggressive after US GAAP              |

Interest expenses = PBO  $\times$  r  
Int  $\downarrow$ , if the employee is young 年轻员工的利率低  
Int  $\uparrow$ , if the plan is mature PBO 下降



## ★ Total Periodic Pension Costs

TPPC: I/S cost + DCI cost, 便于比较不同准则, IFRS, US GAAP TPPC 相等

$$TPPC = PPC \text{ in I/S} + PPC \text{ in OCI} = \Delta F.S - Contribution$$

$$\text{期初 FS} + Contribution - TPPC = \text{期末 FS}$$

## 会计调整

### ① I/S Pension cost 重分类

将 Pension Expense 部分移到 non-operating

US GAAP PPC I/S 调整:

1. + PC 养老金费用
2. - CSC (operating expense)
3. - Interest Expense (同其余 Interest Expense 合并, EBITD)
4. + Actual Return, 无视 G(2), 移到 non-operating (other income)

调整

原本

IS

+ C.S.C → SG&A 原 CSC 在 operating

+ Int Cost → Int Expense

- ECA → Int Income

Amor of P.S.C } non-operating G/L

Amor of A.G/L }

### ② CF 调整. 100% CFO 不合理.

CFF 调动的相反变动 CFO:

缺 D: Contribution < TPPC. 理解为欠钱, 借钱. → CFF+ / CFO-

打款不足

➤ Under-contribution (contribution < TPPC): "Borrowing"

$$✓ \text{ CFF adjustment} = \text{CFF} + (\text{TPPC} - \text{Contribution}) \times (1 - \text{tax rate}) !$$

$$✓ \text{ CFO adjustment} = \text{CFO} - (\text{TPPC} - \text{Contribution}) \times (1 - \text{tax rate})$$

➤ Over-contribution (contribution > TPPC): "Repayment"

$$✓ \text{ CFF adjustment} = \text{CFF} - (\text{Contribution} - \text{TPPC}) \times (1 - \text{tax rate})$$

$$✓ \text{ CFO adjustment} = \text{CFO} + (\text{Contribution} - \text{TPPC}) \times (1 - \text{tax rate})$$

Other post-employment benefits 处理类似于 DB. 更复杂的假设. 假设同 OPB 披露.

Ultimate healthcare trend rate: 长短的 q.

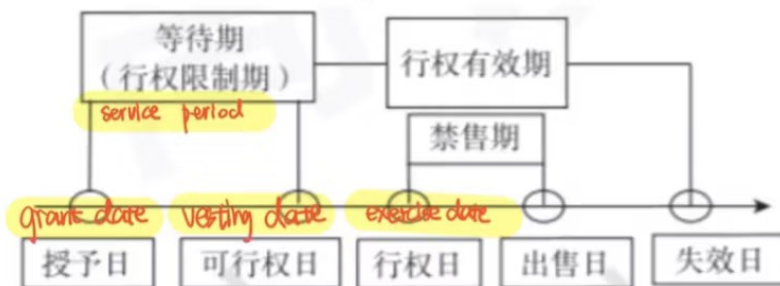
## 股份支付

权结 equity-settled: stock / stock option

现结 cash-settled: stock appreciation rights / phantom shares 模拟股票.

option 变动 → cost 变动 → NI 变动.

cost 在 vesting period 摊销.



## LM3. 跨国

### Transaction 外币交易会计处理

Type of Exposure

外币交易的 exposure: 时间差导致, 期间汇率波动

import purchase: 希望本币升值, 外币贬值. Liability  
export sale: 希望外币升值, 本币贬值. Asset.

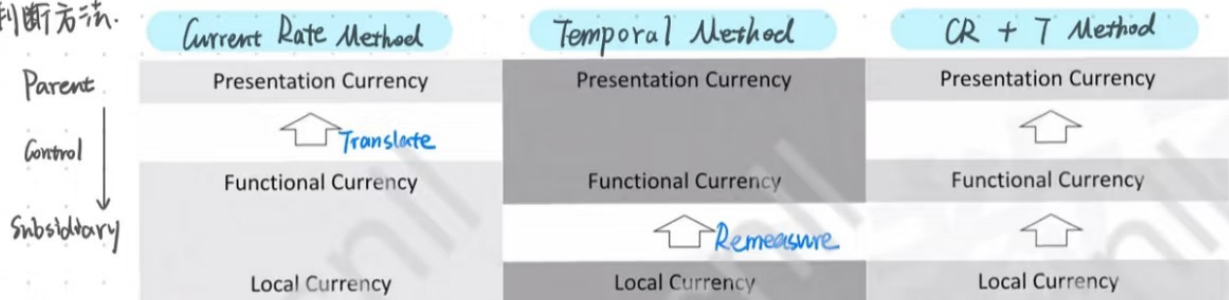
未跨年: 应收, RE → Cash, RE. 计 Realized G/L

跨年: Unrealized G/L 先计在第一年报表. Settle 时与年末比, 计 Realized G/L.

### Translation 外币报表折算

Functional Currency: 关注子公司的 Functional Currency. 销售, 支出, 融资等作为判断依据.

#### 1 判断方法



#### 2 折算

##### Current Rate Method

When the current rate method is used, the cumulative translation adjustment needed to keep the translated balance sheet in balance is reported as a separate component of stockholders' equity

高顿教育  
GOLDEN EDUCATION

|                  |              |
|------------------|--------------|
| Sales Revenue    | Average Rate |
| (COGS)           | Average Rate |
| (SG&A)           | Average Rate |
| Operating Income |              |
| Other Income     | Average Rate |
| EBIT             |              |
| (Interest)       | Average Rate |
| (Tax)            | Average Rate |
| Net Income       | A            |

|              |              |                                   |                           |
|--------------|--------------|-----------------------------------|---------------------------|
| Asset 1      | Current Rate | Liability 1                       | Current Rate              |
| Asset 2      | Current Rate | Liability 2                       | Current Rate              |
| Total Assets |              | Total Liabilities                 |                           |
|              |              | Capital                           | Historical Rate II        |
|              |              | Additional Paid-In                | Historical Rate III       |
|              |              | Retained Earnings                 | RE <sub>beg, PC</sub> + B |
|              |              | Cumulative Translation Adjustment | C                         |
| Total Assets |              | Total Equities                    |                           |

NI - Dividend = ΔRE

A (NI<sub>LC</sub> × Average Rate) - Div<sub>LC</sub> × Historical Rate = B

CTA to balance

先 I/S, 后 B/S

- I/S 折算: Average Rate → NI
- B/S 折算: Asset: current rate  
Liability: current rate.  
Equity: Capital: historical  
A/E  
OCI
- RE<sub>末</sub> = RE<sub>初</sub> + NI - Div
- E - Capital - A/E = OCI

##### Temporal Method

先 B/S, 后 I/S.

inventory & COGS 可能用 weighted average exchange rate, 会与之相关

|                     |                     |                    |                    |
|---------------------|---------------------|--------------------|--------------------|
| Cash                | Current Rate        | Accounts Payable   | Current Rate       |
| Accounts Receivable | Current Rate        | Long-Term Debt     | Current Rate       |
| Inventory           | Historical Rate I   | Total Liabilities  | Sum (↑)            |
| Trading Securities  | Current Rate        | Capital            | Historical Rate IV |
| PPE                 | Historical Rate II  | Additional Paid-In | Historical Rate V  |
| Intangible Assets   | Historical Rate III | Retained Earnings  | 2 A                |
| Total Assets        | Sum (↑)             | Total Equities     | Asset - Liability  |

RE<sub>end</sub> - RE<sub>beg</sub> = ΔRE

A - RE<sub>beg, PC</sub> = B

RE<sub>末</sub> = RE<sub>初</sub> + NI - Div

ΔRE + Dividend = NI

B + Div<sub>LC</sub> × Historical Rate VI = C

\* Assume DC<sub>presentation currency</sub>/FC<sub>local currency</sub>

to balance

5

|                  |                     |
|------------------|---------------------|
| Sales Revenue    | Average Rate        |
| (COGS)           | Historical Rate I   |
| (SG&A)           | Average Rate        |
| (Depreciation)   | Historical Rate II  |
| (Amortization)   | Historical Rate III |
| Operating Income |                     |
| Other Income     | Average Rate        |
| EBIT             |                     |
| (Interest)       | Average Rate        |
| (Tax)            | Average Rate        |
| Net Income       | D                   |

2/NI 主要  
Remeasurement Gains and Losses

Net Income 红书号: 186C39052



## 两种方法对比

| Current Rate Method                                                                                          | Temporal Method                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All assets and liabilities are translated by <b>current rate</b>                                             | "Monetary assets/liabilities" and "non-monetary assets/liabilities measured on the balance sheet date at current value" are translated by <b>current rate</b> |
| All revenues and expenses are translated by <b>average rate</b>                                              | Expenses related with non-monetary items are translated by <b>historical rate</b> : only COGS, Dep, Amo                                                       |
| FIRST income statement<br>THEN balance sheet                                                                 | FIRST balance sheet<br>THEN income statement                                                                                                                  |
| $CTA_0 + \Delta CTA = CTA_1$<br>"Cumulative Translation Adjustment"<br>in the <b>shareholders' equity</b> 存量 | "Remeasurement gains and losses"<br>in the <b>income statement</b> 增量                                                                                         |

### 3 判断 T, G/L

Temporal: Exposure = M.A - M.L. ~ Non-M historical rate 不受影响.  $\begin{cases} >0: \text{Asset. } \uparrow G \downarrow L \\ <0: \text{Liability } \downarrow G \uparrow L \end{cases}$

Current: Exposure = T.A - T.L = Equity.

### 4 T vs C 对指标的影响

- ① 先判断 3 个 rate 大小关系
- ② Ratio 分子, 分母用 Rate 表示.

| LC depreciation (特>A>E)                                       | Temporal                       | Current rate |
|---------------------------------------------------------------|--------------------------------|--------------|
| Current ratio                                                 | Higher                         | Lower        |
| Quick ratio $CA - Inventory / C.L$                            | Same                           | Same         |
| A/R turnover $Sales / A/R$                                    | Same                           | Same         |
| Inventory turnover (LIFO FIFO uncertain)                      | Uncertain                      | Uncertain    |
| Fixed asset turnover $Sales / FA$                             | Lower                          | Higher       |
| Gross profit margin $Sales - COGS / Sales$                    | Lower                          | Higher       |
| Net profit margin, ROE, ROA (Translation gain/loss uncertain) | Uncertain                      | Uncertain    |
| Interest coverage $EBIT / Int. exp$                           | Lower                          | Higher       |
| LTD-to-total capital $LT Debt / D/E$                          | Lower (equity used mixed rate) | Higher       |

### 5 披露

外汇影响 NI 时, translation, transaction 可分开, 可合并披露.  
产生  $\Delta CTA$  时.

### 6 子公司恶性通胀.

标准:  $(1+g)^3 - 1 > 100\% \rightarrow g > 26\%$

IFRS: ① restate for inflation 重述报表, 使购买力保持一致. (Non-M. Equity, I/S items)

B/S: Monetary items 金额固定, 不重述  
Restate:  $x(P_i / P_0)$ , 最后倒算 R/E

I/S: Net G/L in purchasing power to balance, 本质是 M.A 与 M.L 的角逐.  
 $x(P_i / P_{average})$

②  $\times$  Current Exchange Rate

GAAP: ① 用 Temporal Method. (一定不重述报表!)

抗通胀: Asset 存海外资产及 Non-M Asset.

# LM4. Financial Institutions

## Introduction

### 1 Types

- 2 Features { Systemic Importance.  
Heavily Regulated.  
Assets and Liabilities

| Total Capital | Tier 1 Capital | Common Equity Tier 1 Capital | <ul style="list-style-type: none"> <li>common stock</li> <li>issuance surplus related to common stock</li> <li>retained earnings</li> <li>accumulated other comprehensive income</li> <li>certain adjustments including the deduction of intangible assets and deferred tax assets</li> </ul> |
|---------------|----------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                | Other Tier 1 Capital         | other types of instruments issued by the bank that meet certain criteria                                                                                                                                                                                                                      |
|               | Tier 2 Capital |                              | instruments that are subordinate to depositors and to general creditors of the bank, have an original minimum maturity of five years, and meet certain other requirements (期限明确等)                                                                                                             |

## Banks

### ★ CAMELS Approach

1 Capital Adequacy =  $\frac{\text{Capital}}{\text{Risk Weighted Assets}}$

整体 ≥ 8%  
一级 ≥ 6%  
核心一级 ≥ 4.5%

Basel Accord 三大亮点 { Minimum Capital Requirement.  
Minimum Liquidity: LCR 3月 ≥ 100%  
Stable Funding: NSFR 1年 ≥ 100%

### Liquidity Coverage Ratio

$$= \frac{\text{highly liquid assets}}{\text{expected CF}}$$

### Net Stable Funding Ratio

$$= \frac{\text{available stable funding}}{\text{required stable funding}}$$

2 Asset Quality: 定性看 creditworthiness; 定量看 expected loan loss.

3 Management Capabilities

4 Earnings { ① 银行三大收入 { 净利息差, 越稳定越好  
服务 earning  
trading } 相对 stable  
② Fair Value Hierarchy

ROE { unbiased  
sustainable  
sufficient.

5 Liquidity Position: 定量 LCR ≥ 1, NSFR ≥ 1; 定性: Concentration, Mismatch

6 Sensitivity to Market Risk: Interest Rate Risk { hypothetical:  $(A-L) \times r = \text{profit}$   
reality: mismatch → 利率↑

## Insurance

1 P&C 财险: 更保守, 投资的 liquidity 更好  
商业模式: 直销 - 公司人卖  
代销 - 外部人卖  
5 Ratios

✓ Loss and loss adjustment expense ratio = (Loss expense + Loss adjustment expense) / Net premiums earned.

✓ Underwriting expense ratio = Underwriting expense / Net premiums written.

✓ Combined ratio = Loss and loss adjustment expense ratio + Underwriting expense ratio. < 100%

✓ Dividends to policyholders (shareholders) ratio = Dividends to policyholders (shareholders) / Net premiums earned.

✓ Combined ratio after dividends = Combined ratio + Dividends to policyholders (shareholders) ratio.

2 L&I 寿险: 可预测, 相对更激进, liquidity 要求低.

商业模式: 直销 - 公司  
代销 - 个人

|                       | P&C Insurers                               | L&I Insurers                                     |
|-----------------------|--------------------------------------------|--------------------------------------------------|
| Contract Duration     | Short term                                 | Long term                                        |
| Variability of Claims | Claims are more variable                   | Claims are more predictable                      |
| Business Profile      | Protect against loss or damage to property | Provide benefit upon death and a savings vehicle |
| Investment Returns    | Conservative                               | Seek higher returns                              |
| Liquidity             | Requires a high degree of liquidity        | Liquidity was less important to life insurers    |
| Capitalization        | No global standard exists                  | Lower capital requirements                       |



# LM1 Financial Statement Modeling (Sales Based)

高度依赖 Revenue 的预测。

## I/G 预测

方向  
预测

Bottom up / Top down / Hybrid

|                         |
|-------------------------|
| Revenue                 |
| - COGS                  |
| Gross Profit            |
| - SG&A                  |
| Operating Income (EBIT) |
| - Interest Expense      |
| EBT                     |
| - Tax Expense           |
| Net Income              |

Top down: GDP, market, market share 3个 growth 叠加。

Bottom up: Time series:  $Revenue_t = \alpha + \beta \cdot Revenue_{t-1} + \varepsilon$

Return on capital 更多是金融业

Capacity-based measure 更多是零售行业, 单店模式

COGS =  $(1 - \text{Gross Margin}) \times (\text{Future Revenue})$  由 gross margin 倒推。

更精确: 考虑到规模效应提高 Gross Margin; 可分 business 加总。

Fixed Component: 保持不变 or 根据预算

Variable Component: 与 COGS 同比

Net Debt  $\times$  % or Interest Expense - Interest Income 是净值!

EBT  $\times$  Effective Tax Rate %

## B/S 预测

① WC Items

Inventory Turnover =  $\frac{COGS}{(Inv_{初} + Inv_{末})/2}$

等各种 turnover 率决定

② LT Items

投融资基于 budget

③ Equity

Capital: 增发/回购影响。  
R/E: BASE 法则。  
OCI

## Return

$$\left. \begin{aligned} ROIC &= \frac{EBIT(1-t)}{\text{Invested Capital}} = \frac{NOPAT}{\text{Invested Capital (DTE)}} \\ ROCE &= \frac{EBIT}{\text{Capital Employed}} \end{aligned} \right\} \text{税率不同时}$$

## Porter's Five Forces

## Inflation & Deflation

price elasticity of product. 越低越不受影响。

## LT Forecasting

Normalized Earnings.

经济周期。

预测期与投资期. 先精确, 再永续. TV | Historical Multiple Based Approach  
DCF